

Road Damage Detection

A Deep Learning Framework Using UAV Imagery

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Problem Statement

- Manual road inspections are costly and time-consuming
- Safety hazards for inspection personnel
- Limited coverage and periodic checks
- Lack of real-time monitoring capabilities
- Need for efficient automated detection systems

Core Contributions

- Novel Spanish UAV road damage dataset merged with RDD2022
- Transformer-Enhanced YOLO for improved detection
- Comprehensive benchmarking of YOLOv4, v5, v7
- Fully automated drone-based inspection pipeline

System Architecture

Data Acquisition

DJI Air 2S UAV with high-resolution RGB camera

Processing

YOLO-based deep learning models

Output

Real-time damage detection & reporting

Dataset Composition

2,893

Total UAV Images

69.2%

Training Set

6

Damage Classes

Combined Spanish UAV dataset + RDD2022 benchmark (Japan, India, China, Czech Republic, Norway, USA)

Road Damage Classes

- D00: Longitudinal Cracks
- D10: Transverse Cracks
- D20: Alligator Cracks
- D40: Potholes
- Repairs & Block Cracks: Secondary classes

Deep Learning Models Evaluated

YOLOv4: Baseline model with Mosaic augmentation

YOLOv5: Improved architecture with SPADE backbone

YOLOv7: Latest Efficient-YOLO with EfficientNet

YOLOv5+TPH: YOLOv5 with Transformer Prediction Head

Training Configuration

- Hardware: NVIDIA RTX 3090 GPU, 24GB VRAM
- Image Resolution: 640 × 640 pixels
- Data Augmentation: Flip, rotate ($\pm 15^\circ$), brightness/contrast adjust
- Training Platform: Roboflow for dataset organization
- Framework: PyTorch with official YOLO implementations

YOLOv7: Best Performance

73.7%

mAP@0.5

0.788

Precision

0.714

Recall

11.4ms

Inference Time

Model Comparison: mAP@0.5

YOLOv4:

26.8%

YOLOv5:

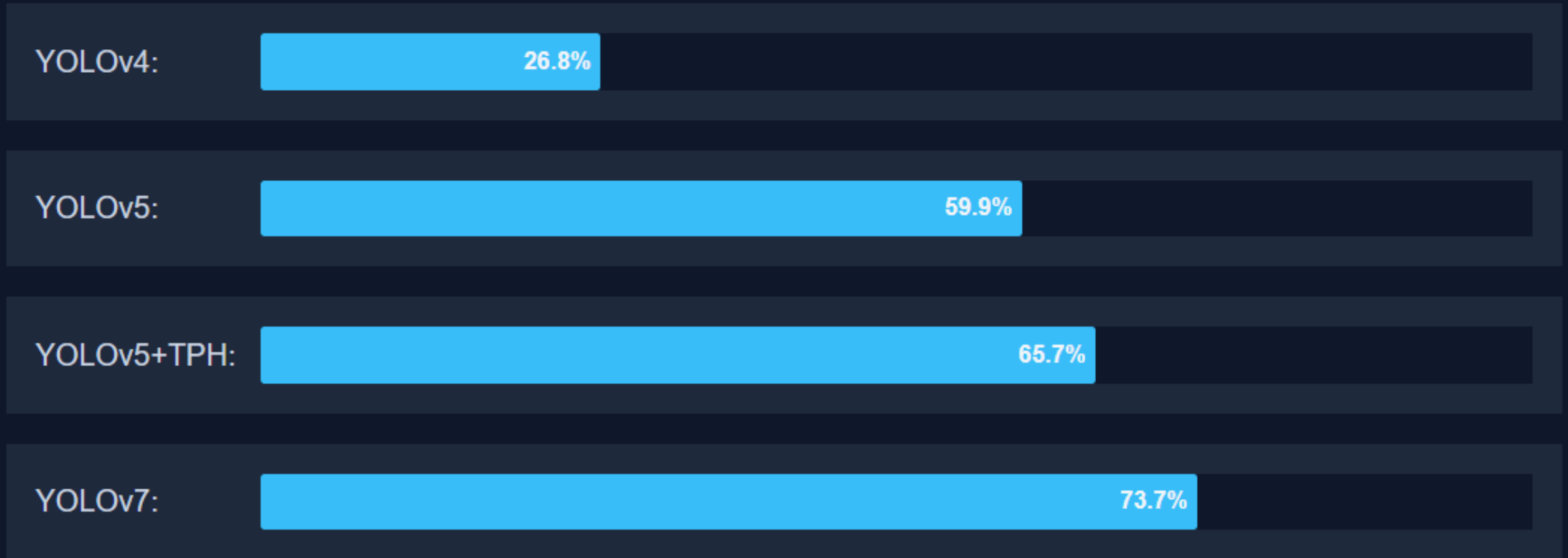
59.9%

YOLOv5+TPH:

65.7%

YOLOv7:

73.7%



YOLOv5 + Transformer Results

65.7%

mAP@0.5

6.1ms

Inference Time

Enhanced spatial feature modeling with self-attention mechanisms for improved localization in complex scenes

Key Findings

- YOLOv7 achieves highest accuracy (73.7% mAP@0.5)
- Real-time inference enables field deployment (11.4ms)
- Transformer head improves handling of scale variations
- Exceptional performance on repairs (82.7% mAP@0.5)
- Multi-country dataset improves model generalization

Challenges & Solutions

Scale Variation: Addressed with Transformer Prediction Head

Class Imbalance: Mitigated via data augmentation techniques

Real-time Performance: YOLOv7 achieves 87 FPS on GPU

Real-World Applications

- Automated highway and urban road monitoring
- Infrastructure maintenance planning
- Cost reduction in inspection operations
- Safety improvement for personnel
- Large-scale deployment capability (87 FPS)

Future Directions

- Integration of multispectral and LiDAR data
- Lightweight models for edge device deployment
- Severity-level damage assessment framework
- Real-time onboard processing on drones
- Extended geographic dataset collection



Thank you